Slide 1: Executive Overview -- Deterministic Physics First

* Mission: Govern admission (D1), physical placement (D2), live conflict arbitration (D3), graceful degradation (D4), multi-crew scheduling (D5: 14/6/0), and fail-closed physical accounting (D6) across 8 coupled physical resource classes (power, cooling, positions, mast, mass_stability, data, compute, time) on a 72 m Software-Defined Vessel.

* Core Principle: Hard physical and life/vehicle safety limits are satisfied first; mission capability score, free margin, and displacement cost are optimized second. No priority weight can ever override a hard physical limit or drone ditching deadline.

* Zero Opaque Runtime AI: 100% deterministic, inspectable math (exact cooling curve, component-level power/water/air splits, parameterized witness explanations, 1 Hz internal simulation stepping).

Slide 2: Fail-Closed Validation & Declarative Guest Ontology (D1)

* Strict Schema & Type Discipline (B2, F3, H06, H13, R2, R4):
  * Rejects NaN, Infinity, negative or zero budgets, booleans used as numbers (True != 1), string booleans ("false"), unknown YAML/HTTP JSON fields (UNKNOWN_FIELD), dependency cycles, and invalid cooling coupling (cooling_water_w + cooling_air_mast_w == sustained_power_w).
  * Reproduces and defeats all 10 adversarial input flaws (resource_engine_repro.json) and all 28 verifier mutation classes (false_accepts = 0).
* Six Declarative Guest Cards (G1-G6) with Field-Level Provenance:
  * Official G1 (45 kg, 1.5 kW air on mast_3), G2 (3,500 kg, 20/6 kW), G3 (5,500 kg, 25/1 kW), G4 (140+850 = 990 kg per GRAZIANO_ROOF_AND_G4_MASS_0904, 35/10/2 kW), G5 (3,000+25 = 3,025 kg, 45/8 kW), plus custom civil card G6 (420 kg, port_only=true, all fields ASSUMED) and zero-code dynamic G7.

Slide 3: Scenario A (D1/D2) -- 8-Class Admission, Roof Clearance & Errata Proof

* Organizer Clarifications Verified (GRAZIANO_B_COMPOSITION_0855, GRAZIANO_ROOF_AND_G4_MASS_0904):
  * With G4 = 990 kg, G3+G4+G5 = 9,515 kg ~= 9.52 t and G1+G3+G4+G5 = 9,560 kg (440 kg mass margin, 9,853.0 kg-eq stability with G5 antenna on hangar_roof, 2,147.0 kg-eq margin).
  * Across C(5,4)=5 sets, 3 exceed 10,000 kg ({G1,G2,G3,G4}=10,035 kg, {G1,G2,G3,G5}=12,070 kg, {G2,G3,G4,G5}=13,015 kg), while {G1,G2,G4,G5}=7,560 kg and {G1,G3,G4,G5}=9,560 kg require G5_MAST_ANTENNA on hangar_roof under conditional policy ROOF_OUTSIDE_G4_RECOVERY_V1 (YES_IF until verified; NO under CONFLICT policy).
* Single Platform Modification Counterfactual (mast_4):
  * Adding mast_4 (<= 60 kg, <= 0.6 m2, <= 2,000 W air) separates G1 (1,500 W on mast_3) and G5 antenna (1,200 W on mast_4, 9,898.0 kg-eq), removing the roof-clearance condition while mass/bays/stability still block the other three 4-guest sets.

Slide 4: Scenario B (D3) -- Installed G1+G3+G4+G5 & Causal Acoustic Handshake

* Active 4-Guest Set (G1+G3+G4+G5):
  * G1 on mast_3 (45 kg, 1.5 kW air, LINE_A_FWD), G3 in bay1 (5,500 kg, 1.0 kW survey / 25.0 kW firm charge, LINE_A_FWD), G4 in hangar/flight_deck (990 kg, 10.0 kW patrol / 35.0 kW recovery, LINE_B_AFT), G5 container in bay2 + antenna on hangar_roof (3,025 kg, 45.0 kW = 43.8 kW water + 1.2 kW roof air, LINE_B_AFT). G2 is not installed in Scenario B.
* Causal Command-ACK & A.6 Transition Timing:
  * At T=480 s, G4 reports 18% battery (720 s deadline -> T <= 1,200 s) and D3 reserves 35.0 kW (82.5 / 106.5 kW actual/committed).
  * Turning into 22 kn wind requires verified ACOUSTIC_HOLD_CONFIRMED ACK from G3 (15 s nominal, [10, 60] s window), 90 s turn/stabilization into wind, and 240 s approach/landing: rescue_primary lands at T=855 s (345 s margin); survey_primary lands at T=1,095 s (105 s margin).

Slide 5: Scenario B (D4) -- Graceful Degradation, Feeders & Water/Air Split

* Component-Level Power, Feeder & Thermal Ledger (G1+G3+G4+G5):
  * Electrical Power (actual / committed): 57.5 / 81.5 kW initial -> 82.5 / 106.5 kW with G4 35 kW reservation -> when generator reserve drops 200 kW -> 100 kW at T=720 s, 106.5 kW > 100 kW (6.5 kW committed deficit) triggers G5 step-down (45.0 -> 8.0 kW), yielding 45.5 / 69.5 kW <= 100 kW (30.5 kW committed margin) and 0 infeasible ticks (91/91)!
  * Chilled-Water & Air Domains: Water load goes 54.8 / 78.8 kW -> 79.8 / 103.8 kW -> 43.5 / 67.5 kW <= 72 kW after G5 step-down (43.8 -> 7.5 kW water), surviving 30 deg C + -40% loop loss (72 kW). Air cooling is 1,500 W (mast_3) + 1,200 W (hangar_roof) = 2,700 W (2,000 W degraded).
  * Feeder Routing (<= 150 kW/line): LINE_A_FWD = 2.5 / 26.5 kW; LINE_B_AFT = 80.0 -> 43.0 kW.

Slide 6: Scenario C (D5) -- Crew Regimes (14 / 6 / 0) & Implicit Timing Rules

* Concrete Implicit Rule (Trigger -> Effect -> Confirmation):
  * Urgent safety actions (STOP, HOLD, SECURE <= 200 ms, < 1 ms measured) execute autonomously across 14 / 6 / 0.
  * Non-urgent confirmation & flight-deck clearing scale with crew dial: 14 crew (9 deck avail, 30 s confirm, 240 s deck clear; A' replacement permitted at T+1260 s), 6 crew (2 deck avail, 120 s confirm, 600 s deck clear; A' replacement blocked), 0 crew (0 avail; software mode changes automated, manual evolutions and occupied-deck recovery PROHIBITED).
* Component-Split Overconsumption (+20% G5, ?6.5):
  * If G5 surges +20% to 54.0 kW electrical (52.8 kW water + 1.2 kW roof air), D4 immediately throttles G5 to 8.0 kW (7.5 kW water + 0.5 kW air).

Slide 7: Physical Account (D6) -- 8 Assumption Families & Code Traceability

* Complete 1-Page Physical Account (docs/PHYSICAL_ACCOUNT_D6.pdf):
  * Maps all 8 assumption families (Provenance/mass, Roof & drone clearance, Air & component split, Power & feeders/reservations, Chilled water & thermal inertia, State & A.6 transition timing, Compute rack & data links, Priorities & degraded capability) to exact YAML files and file.py::symbol_name functions.
  * Enforces 4 physical compute nodes (node_1..node_4, 32c / 128 GiB / 1 acc each; 80% allocatable = 25c / 102.4 GiB per node, 3 assignable discrete accelerators across cluster).

Slide 8: Operator Interface -- Synchronized 2D Console & 3D Vessel Twin

* Authoritative Shared Session (server.py + ui/ + viz3d/):
  * Both the single-screen 2D Operator Console (http://127.0.0.1:8765) and the Three.js 3D Vessel Twin (viz3d/) bind to the same GovernorSession (/api/state, /api/advance, /api/perturb, /api/admit_live).
  * Zero Silent Synthetic Fallback: If the backend disconnects in live mode, the 3D Twin displays an explicit STALE / DESCONECTADO banner rather than fabricating synthetic telemetry.

Slide 9: Dual-Platform Architecture & Live Ventuno Q Parity

* Decoupled Laptop + Edge Board Execution (B1, DEP-01..03):
  * MacBook Pro (arm64, Python 3.13.15): Full test suite, 1 Hz event-driven simulator, 90 s (20x) compressed demo, and 2D/3D Operator Interface.
  * Arduino Ventuno Q (training06, Qualcomm QCS8300 Monza aarch64, 14 GiB RAM, Ubuntu 24.04.4): Live SSH deployment at /home/arduino/sdv-resource-governor.
  * Bit-for-Bit Semantic Parity: Identical campaign_semantic_sha256 verified across arm64 macOS and live aarch64 Ventuno Q, with 28 / 28 adversarial mutations rejected (false_accepts = 0).

Slide 10: 90-Second Demo Script & Honest Operational Boundaries

* 90-Second (20x) Compressed Mission Walkthrough (docs/DEMO_SCRIPT_90S.md):
  * 0-20 s: Port Admission (G1+G3+G4+G5 = 9,560 kg, 9,853 kg-eq, ROOF_OUTSIDE_G4_RECOVERY_V1 verified) + zero-code G7.
  * 20-55 s: Live Scenario B D3 arbitration (T=480 s G4 18% battery -> G3 ACOUSTIC_HOLD ACK -> turn into wind -> 855 s / 1,095 s touchdown).
  * 55-90 s: T=720 s generator loss (106.5 -> 69.5 kW power, 103.8 -> 67.5 kW <= 72 kW water via G5 step-down) + 14/6/0 crew dial.
* Explicit Rehearsal & Boundary Disclosure:
  * Software command dispatch is < 1 ms (<= 200 ms budget); physical breaker/winch/turning durations follow Annex A.6; live human spoken 90-second rehearsal status is explicitly recorded as DEFERRED_OWNER.